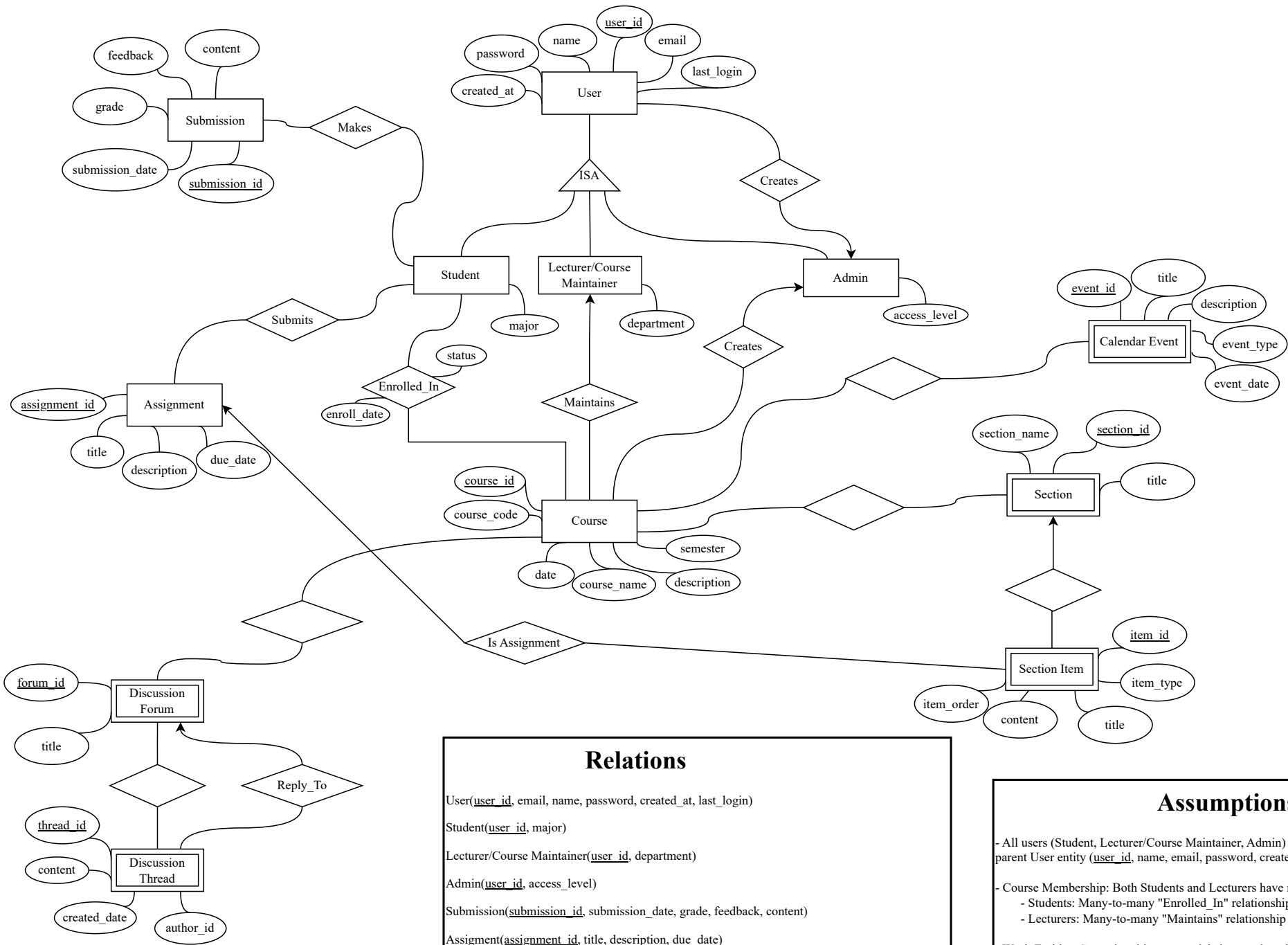




ID#: 620162698

Relations

User(user_id, email, name, password, created_at, last_login)
Student(user_id, major)
Lecturer/Course Maintainer(user_id, department)
Admin(user_id, access_level)
Submission(submission_id, submission_date, grade, feedback, content)
Assignment(assignment_id, title, description, due_date)
Course(course_id, course_code, date, course_name, description, semester)
Calendar Event(event_id, title, description, event_type, event_date)
Section(course_id, section_id, section_name, title)
Section Item(section_id, item_id, item_type, title, content, item_order)
Discussion Forum(course_id, forum_id, title)
Discussion Thread(forum_id, thread_id, content, created_date, authour_id)
Makes(user_id, submission_id)
Submits(assignment_id, user_id)
Enrolled_In(user_id, course_id, status, enroll_date)
Maintains(user_id, course_id)
Creates(course_id, user_id)
Creates(user_id)
Is_Assignment(item_id, assignment_id)
Reply_To(thread_id, forum_id)

Assumptions

- All users (Student, Lecturer/Course Maintainer, Admin) inherit common attributes from a parent User entity (user_id, name, email, password, created_date, last_login).
 - Course Membership: Both Students and Lecturers have relationships with Course:
 - Students: Many-to-many "Enrolled_In" relationship
 - Lecturers: Many-to-many "Maintains" relationship
 - Weak Entities: Several entities are modeled as weak entities:
 - Discussion Forum: Exists only within a Course
 - Discussion Thread: Exists only within a Forum
 - Calendar Event: Exists only within a Course
 - Section: Exists only within a Course
 - Section Item: Exists only within a Section
- Discussion Thread Replies: Modeled as a recursive relationship "Reply_To" on Discussion Thread. A thread can reply to one parent thread, and a thread can have many replies.
- Assignment as Section Item: Assignments are modeled as a specialization/type of Section Item through the "Is_Assignment" relationship. This allows assignments to appear in the course content structure while maintaining separate assignment-specific attributes. On the diagram, I have the cardinality as One-to-Many but I believe it best matches a 0 .. 1 from Section Item (not all items are assignments) and 1 from Assignment (every assignment must be a section item).
 - Calendar Events: One-to-many relationship between Course and Calendar Event. Events are course-specific and managed by course maintainers.
 - Admin Creating Users: Modeled as a "Creates" relationship from Admin to User (one admin can create many users).
 - Admin Creating Courses: Modeled as a separate "Creates" relationship from Admin to Course.

Constraints not captured in ERD

Only Course Maintainers assigned to a course can create/modify calendar events for that course.
Only Course Maintainers assigned to a course can add/update/remove section items for that course.
Students can only view content for courses they're enrolled in.
Students can only create threads in forums of courses they're enrolled in.
Only the original author or course maintainer can edit/delete a thread.

Submissions can only be made before the assignment due date (or with late penalty).
Students cannot enroll after the course enrollment deadline.
Calendar events must have future dates when created.
Enrollment_date must be before or equal to the course start date.

Email must be in valid email format and unique across all users.
Password must meet minimum security requirements.

A student cannot submit more than once to the same assignment.
A course must have at least one course maintainer at all times.
Cannot delete a section if it contains section items.

Additional attributes

The added attributes are as follows: User.created_date, User.last_login, Student.major, Lecturer.department, Admin.access_level, Course.course_code, Course.semester & year, Enrolled_In.enrollment_date, Enrolled_In.status, Discussion_Thread.created_date, Calendar_Event.event_type, Section.section_order, Section_Item.item_type, Section_Item.item_order, Submission.grade, Submission.feedback.

I believe these attributes are standard in learning management systems like Moodle for instance and are essential for:

- Proper academic record keeping.
- User experience.
- Administrative functions,
- Security and auditing.